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SUPERSEDING
Army 51-73-42
7 July 1944

MILITARY SPECIFICATION

INDICATOR, REGULATOR, M1

This specification consists of this cover sheet and Specification 51-73-42, dated 7 July 1944, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

Custodian:
Army - Ordnance Corps

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U. S. ARMY
SPECIFICATION

No. 51-73-42
7 July 1944

INDICATOR, REGULATOR, M1

A. APPLICABLE SPECIFICATIONS.

A-1. Ordnance Department drawing D46044, titled "Indicator, Regulator, M1", lists or indicates all drawings which form a part of this specification.

A-2. All drawings and specifications shall be of the issue in effect on date of invitation for bids. The issue in effect shall be that issue which the contracting officer shall designate or furnish to the bidder.

B. TYPE.

B-1. This specification covers one type of indicator known as "Indicator, Regulator, M1".

C. MATERIAL AND WORKMANSHIP.

C-1. Material and workmanship shall be in accordance with the drawings, general specifications, and material specifications forming parts of this specification.

C-2. Commercial material. Components and materials of commercial nature for which no requirements are contained herein or for which no specifications are listed on the list of specifications drawings, shall be of a type and quality most suitable for the purpose.

C-3. Insulated wire used in the fabrication of the instrument shall be in accordance with Frankford Arsenal Specification FXS-599, Wire, Single Conductor Apparatus, Synthetic Compound Insulated.

D. GENERAL REQUIREMENTS.

D-1. See U. S. Army Specification No. 51-7, "Instruments, Fire Control, General Specification Governing the Manufacture and Inspection of".

E. DETAIL REQUIREMENTS.

E-1. See detail drawings forming part of this specification.

E-2. Adjustments.

E-2a. The electrical indexes of the coarse and fine repeaters shall be opposite the zero graduation of the coarse and fine dials, respectively, when the repeaters are electrically zeroed and when the coarse and fine repeater and synchro-transformer external adjusting screws have the same amount of movement, in each direction, to their stops. The electrical zero position is defined in Frankford Arsenal Tentative Specification FXS-348, A-C Synchronous Units. (In those indicators where terminal 21 is connected to terminal 3-B instead of to terminal 21 on the demodulator, it is necessary to run a temporary jumper from 3-B on terminal board to 21 on the demodulator in order to make electrical tests.)

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E-2b. The synchronizer roller shall be electrically centered on the insulated segment of the synchronizer when the coarse repeater is at the electrical zero position.

E-2c. The lagmeter shall indicate zero with the power "off".

E-2d. When the indicator regulator is properly connected electrically to two external standard transmitters geared together, ratio 16 to 1, (or equivalent dummy director) and set to zero; the coarse and fine indicator dials shall be at zero, the synchronizer roller in the center of the insulated segment, and the synchro-transformer set to zero voltage as checked by volmeter or oscilloscope.

E-2d(1). Rotate the standard fine transmitter in increments of 400 mils for 6400 mils rotation of the standard coarse transmitter. The error introduced by mechanical drag on the coarse indicator regulator dial shall not exceed 8 mils as measured on the dial of the fine standard transmitter.

E-2d(2). During the operation specified in the preceding paragraph, the lagmeter shall show a clockwise deflection up to 200 mils when the standard fine transmitter is rotated from zero in a clockwise rotation and the mechanical train is kept stationary.

E-2d(3). With the indicator regulator set to zero, five to six mils clockwise movement of the mechanical dial, produced by turning the differential shaft, shall deflect the lagmeter pointer counter-clockwise to a position opposite the dot on the face of the meter. Reversing the operation shall result in a clockwise deflection of the lagmeter to a position opposite the other dot on the face of the meter.

E-3. Backlash from the input shaft to the fine mechanical index shall not exceed 1/4 mil at any setting as measured on the fine dial.

E-4. All moving parts shall function properly without irregularities, undue friction or looseness.

E-5. Phase relationship between terminals 24 and 25, when determined by a cathode ray oscillograph connected across S_1 and S_2 (Ordnance drawing C78742) through the medium of an insulating transformer, shall be the same as the phase relationship of the fine synchro-transformer.

E-6. Insulation Breakdown and Resistance.

E-6a. Insulation breakdown shall not occur when 1250 volts (rms), 60 cycles, a-c, is applied for one minute between ungrounded insulated electrical circuits and ground, and between isolated circuits. The peak voltage shall not exceed 1.41 times the effective voltage.

E-6b. Insulation resistance between ungrounded insulated electrical circuits and ground, and between isolated circuits, shall be not less than 50 megohms

E-7. Soldered electrical connections. Copper wires and lugs shall have a tinned finish prior to soldering. Only rosin flux shall be used. After soldering the joint shall be carefully washed with alcohol to remove all traces of rosin. The joint shall then be coated with an electrical insulating lacquer based upon an alkyd rosin. At least 75 percent of the contact area of soldered joints shall be properly fused.

E-8. Operating temperature. The indicator regulator shall function satisfactorily throughout an ambient temperature range of minus 45° F. to plus 150° F.

E-9. Lubrication shall be in accordance with U. S. Army Specification No. 51-7 and/or instructions from the contracting officer.

E-10. Manufacture of extra parts shall be in accordance with the drawings and this specification.

F. METHODS OF INSPECTION AND TESTS

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F-1. Each instrument, including extra parts shall be inspected for conformance with section E of this specification, and shall be subject to inspection during process of manufacture and assembly as specified in U. S. Army Specification No. 51-7.

F-2. Insulation breakdown test. The applied voltage shall be raised slowly to the value, specified in paragraph E-6a, and held at that value for a period of one minute.

F-3. The insulation resistance test shall be made after the insulation breakdown test. The resistance shall be measured while 500 volts d-c are applied.

F-4. Temperature Test. Three out of the first ten regulators manufactured under the contract, and at least two per cent of the remaining regulators taken at uniform intervals throughout the balance of the contract, shall be selected by the inspector for test at both the high and low extremes of the temperature range specified in paragraph E-8. The regulators shall be allowed to soak at each temperature for five hours immediately prior to test. Failure of any regulator to conform to the functional requirements specified in section E at those temperatures and subsequently at room temperature (approximately 70° F.) shall, at the discretion of the inspector, be cause for tests of additional regulators, and correction of all faulty regulators by the contractor.

F-5. Rain test.- One instrument out of every 100, as selected by the inspector, shall be given the artificial rain test as specified in section V of U. S. Army Specification No. 51-7.

G. PACKING AND MARKING

G-1. Packing

G-1a. Packaging and packing for domestic shipment shall conform with the instructions forming part of the contract or furnished by the contracting officer in connection with the contract.

G-1b. Packaging and packing for export shipment shall comply with the applicable requirements of U. S. Army Specification No. 100-14, and/or instructions furnished by the contracting officer in connection with the contract.

G-2. Marking for shipment shall be in accordance with U. S. Army Specification No. 100-2, "Standard Specification for Marking Shipments".

H. NOTES

H-1. Extra parts and equipment - See section VII, of U. S. Army Specification No. 51-7.

H-2. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

NOTICE: This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Note. Copies of this specification may be obtained from the Chief of Ordnance, Army Service Forces, Washington, D. C.

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